

Tuple Creation

```
In [533]: tup1 = ()      # Empty tuple
```

```
In [534]: tup2 = (10,30,60)      # tuple of integers numbers
```

```
In [535]: tup3 = (10.77,30.66,60.89)      # tuple of float numbers
```

```
In [536]: tup4 = ('one','two' , "three")  # tuple of strings
```

```
In [537]: tup5 = ('Asif', 25 ,(50, 100),(150, 90))  # Nested tuples
```

```
In [538]: tup6 = (100, 'Asif', 17.765)  # Tuple of mixed data types
```

```
In [539]: tup7 = ('Asif', 25 ,[50, 100],[150, 90] , {'John' , 'David'} , (99,22,33))
```

```
In [540]: len(tup7) #Length of List
```

```
Out[540]: 6
```

Tuple Indexing

```
In [541]: tup2[0] # Retrieve first element of the tuple
```

```
Out[541]: 10
```

```
In [542]: tup4[0] # Retrieve first element of the tuple
```

```
Out[542]: 'one'
```

```
In [543]: tup4[0][0] # Nested indexing - Access the first character of the first tuple element
```

```
Out[543]: 'o'
```

```
In [544]: tup4[-1] # Last item of the tuple
```

```
Out[544]: 'three'
```

```
In [545]: tup5[-1] # Last item of the tuple
```

```
Out[545]: (150, 90)
```

Tuple Slicing

```
In [560]: mytuple = ('one' , 'two' , 'three' , 'four' , 'five' , 'six' , 'seven' , 'eight')
```

```
In [547]: mytuple[0:3] # Return all items from 0th to 3rd index Location excluding the item at 3rd index
```

```
Out[547]: ('one', 'two', 'three')
```

```
In [548]: mytuple[2:5] # List all items from 2nd to 5th index Location excluding the item at 5th index
```

```
Out[548]: ('three', 'four', 'five')
```

```
In [549]: mytuple[:3] # Return first three items
```

```
Out[549]: ('one', 'two', 'three')
```

```
In [550]: mytuple[:2] # Return first two items
```

```
Out[550]: ('one', 'two')
```

```
In [551]: mytuple[-3:] # Return Last three items
```

```
Out[551]: ('six', 'seven', 'eight')
```

```
In [552]: mytuple[-2:] # Return Last two items
```

```
Out[552]: ('seven', 'eight')
```

```
In [553]: mytuple[-1] # Return Last item of the tuple
```

```
Out[553]: 'eight'
```

```
In [554]: mytuple[:] # Return whole tuple
```

```
Out[554]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

Remove & Change Items

```
In [555]: mytuple
```

```
Out[555]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [556]: del mytuple[0] # Tuples are immutable which means we can't DELETE tuple items
```

```
-----  
TypeError                                Traceback (most recent call last)  
<ipython-input-556-667a276aa503> in <module>  
----> 1 del mytuple[0]  
  
TypeError: 'tuple' object doesn't support item deletion
```

```
In [557]: mytuple[0] = 1 # Tuples are immutable which means we can't CHANGE tuple items
```

```
-----  
TypeError                                Traceback (most recent call last)  
<ipython-input-557-4cf492702bfd> in <module>  
----> 1 mytuple[0] = 1  
  
TypeError: 'tuple' object does not support item assignment
```

```
In [561]: del mytuple # Deleting entire tuple object is possible
```

Loop through a tuple

```
In [570]: mytuple
```

```
Out[570]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [571]: for i in mytuple:  
          print(i)
```

```
one  
two  
three  
four  
five  
six  
seven  
eight
```

```
In [572]: for i in enumerate(mytuple):  
          print(i)
```

```
(0, 'one')  
(1, 'two')  
(2, 'three')
```

Tuple Membership

```
In [577]: mytuple
```

```
Out[577]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [578]: 'one' in mytuple # Check if 'one' exist in the list
```

```
Out[578]: True
```

```
In [579]: 'ten' in mytuple # Check if 'ten' exist in the list
```

```
Out[579]: False
```

```
In [581]: if 'three' in mytuple: # Check if 'three' exist in the list
          print('Three is present in the tuple')
          else:
          print('Three is not present in the tuple')
```

Three is present in the tuple

```
In [583]: if 'eleven' in mytuple: # Check if 'eleven' exist in the list
          print('eleven is present in the tuple')
          else:
          print('eleven is not present in the tuple')
```

eleven is not present in the tuple

Index Position

```
In [586]: mytuple
```

```
Out[586]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [587]: mytuple.index('one') # Index of first element equal to 'one'
```

```
Out[587]: 0
```

```
In [586]: mytuple
```

```
Out[586]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [587]: mytuple.index('one') # Index of first element equal to 'one'
```

```
Out[587]: 0
```

```
In [590]: mytuple.index('five') # Index of first element equal to 'five'
```

```
Out[590]: 4
```

```
In [591]: mytuple1
```

```
Out[591]: ('one', 'two', 'three', 'four', 'one', 'one', 'two', 'three')
```

```
In [593]: mytuple1.index('one') # Index of first element equal to 'one'
```

```
Out[593]: 0
```

Sorting

```
In [594]: mytuple2 = (43,67,99,12,6,90,67)
```

```
In [595]: sorted(mytuple2) # Returns a new sorted list and doesn't change original tuple
```

```
Out[595]: [6, 12, 43, 67, 67, 90, 99]
```

```
In [596]: sorted(mytuple2, reverse=True) # Sort in descending order
```

```
Out[596]: [99, 90, 67, 67, 43, 12, 6]
```